Week 1 (SIT1001 Unit 1 Part 2 - 2 hours)

1.3 CONDITIONAL PROBABILITY

We introduce the idea of conditional probability by means of an example.

EXAMPLE 1.12

Suppose that we are given 20 tulip bulbs that are similar in appearance and told that 8 will bloom early, 12 will bloom late, 13 will be red, and 7 will be yellow, in accordance with the various combinations as shown in Table below.

	Early (E)	Late (L)	Totals
Red (R)	5	8	13
Yellow (Y)	3	4	7
Totals	8	12	20

Assume that each tulip bulb is 'equally likely' to be selected.

If one bulb is selected at random, the probability that it will produce a red tulip (R) is ___.

Suppose, however, that close examination of the bulb will reveal whether it will bloom early *(E)* or late *(L)*.

If we **consider an outcome only if it results in a tulip bulb that will bloom early**, only eight outcomes in the sample space are now of interest. Thus, under this limitation, what is the natural probability value to be assigned to *R*? In other word, $P(R|E)$ = ___,

$P(R|E)$ is read as the probability of *R* given that *E* has occurred.

Note that

$$P(R|E) = \frac{n(R \cap E)}{n(E)} = \frac{n(R \cap E)/20}{n(E)/20} = \frac{P(R \cap E)}{P(E)}$$

where $n(R \cap E)$ and $n(E)$ are the numbers of outcomes in events $R \cap E$ and E respectively ■

Example 1.12 indicates that, in some random experiments, we are interested only in those outcomes which are **elements of a subset *B* of the sample space S**.

This means, for our purposes, that the sample space is effectively the subset *B*.

We are now confronted with the problem of **defining a probability set function with *B*** as the "new" sample space. That is, for a given event *A*, we want to define $P(A|B)$, the probability of *A*, considering only those outcomes of the random experiment that are elements of *B*.

DEFINITION 1.2

The conditional probability of an event *A*, given that event *B* has occurred, is defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

provided that $P(B) > 0$.

EXAMPLE 1.13 (Example 1.4-2, HT)

If $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, then

$$P(A|B) = 0.3/0.5 = 0.6;$$

$$P(B|A) = P(A \cap B) \,/\, P(A) = 0.3/0.4 = 0.75$$ ■

EXAMPLE 1.14 (Example 1.4-3, HT)

Suppose that $P(A) = 0.7$, $P(B) = 0.3$, and $P(A \cap B) = 0.2$. Given that the outcome of the experiment belongs to *B,* what then is the probability of *A?*

■

EXAMPLE 1.15 (Example 1.4-4, HT)

A pair of four-sided dice is rolled and the sum is determined. Let *A* be the event that a sum of 3 is rolled, and let *B* be the event that a sum of 3 or a sum of 5 is rolled. Find the probability that a sum of 3 is rolled before a sum of 5 is rolled.

■

It is interesting to note that conditional probability satisfies the axioms for a probability function, namely, with $P(B) > 0$,

a. $P(A|B) \geq 0$,

b. $P(B|B) = 1$,

c. If A_1,A_2, A_3,... are events and $A_i \cap A_j = \emptyset, i \neq j$, then

$$P(A_1 \cup A_2 \cup \cdots \cup A_k|B) = P(A_1|B) + P(A_2|B) + \cdots + P(A_k|B)$$

for each positive integer k, and

$$P(A_1 \cup A_2 \cup A_3 \cup \cdots|B) = P(A_1|B) + P(A_2|B) + P(A_3|B) + \cdots$$

for an infinite, but countable, number of events.

DEFINITION 1.3 (Definition 1.4-2, HT)

The probability that two events, A and B, both occur is given by the **multiplication rule**,

$$P(A \cap B) = P(A)P(B|A)$$

or by

$$P(A \cap B) = P(B)P(A|B)$$

EXAMPLE 1.16 (Example 1.4-6,HT)

A bowl contains seven blue chips and three red chips. Two chips are to be drawn successively at random and without replacement. Compute the probability that the first draw result in a red chip (A) and the second draw results in a blue chip (B).

■

EXAMPLE 1.17 (xample 1.4-7, HT)

From ordinary deck of playing cards, cards are to be drawn successively at random and without replacement. Find the probability that the third spade appears on the sixth draw.

■

EXAMPLE 1.18 (Example 1.4-10, HT)

A grade school boy has five blue and four white marbles in his left pocket and four blue and five white marbles in his right pocket. If he tranfers one marble at random from his left to his right, what is the probability of his than drawing a blue marble from his right pocket? For notation, let *BL, BR*, and *WL* denote drawing blue from left pocket, blue from right pocket, and white from left pocket, respectively.

■

EXAMPLE 1.19 (Example 1.4-11, HT)

An insurance company sells a number of different policies, among which 60% are for autos, 40% are for homeowners, and 20% are for both types of policy. Let A_1 be people with only an auto policy, A_2 people with only a homeowner policy, A_3 people with both types of policy, and A_4 those with only other types of policy.

(a) If a person is selected at random from the policyholders, then determine $P(A_1)$, $P(A_2)$, $P(A_3)$ and $P(A_4)$.

(b) Let B be the event that are policyholder will renew at least one of the auto or homeowner policies. Say from past experience that we can assign the conditional probabilities $P(B|A_1) = 0.6$, $P(B|A_2) = 0.7$, and $P(B|A_3) = 0.8$. Given that the person selected at random has an auto or homeowner policy, what is the conditional probability that the person will renew at least one of those policies?

■

1.4 INDEPENDENT EVENTS

For certain pairs of event, the occurrence of one of them may or may not change the probability of the occurrence of the other. In the latter case, they are said to be **independent events**.

EXAMPLE 1.19 (byjus.com/maths/independent-events)

Consider an example of rolling a die. If A is the event 'the number appearing is odd' and B be the event 'the number appearing is a multiple of 3', then

P(A) =

P(B) =

Also A and B is the event 'the number appearing is odd and a multiple of 3' so that

$P(A \cap B)$ =

Now, we can calculate

$$P(A|B) = \frac{P(A \cap B)}{P(B)} =$$

Comment:

P(A) = $P(A|B)$ =

This implies that ccurrence of event B **has not affected** the probability of occurrence of the event A..

Using the Multiplication rule of probability, P(A ∩ B) = P(B) $P(A|B)$. That is, if A and B are independent events, then $P(A|B) = P(A)$

Using the Multiplication rule of probability, P(A ∩ B) = $P(B)\ P(A|B) = P(B)\ .P(A)$

DEFINITION 1.4 (Definition 1.5-1, HT)

Events A and B are **independent** if and only if $P(A \cap B) = P(A)P(B)$. Otherwise A and B are called **dependent** events.

(Example 1.5-2, HT)

A red die and a white die are rolled. Let event $A = \{4\ on\ the\ red\ die\}$ and event $B = \{sum\ of\ dice\ is\ odd\}$. Of the 36 equally likely outcomes, 6 are favorable to A, 18 are favorable to B, and 3 are favorable to $A \cap B$. Show that A and B are independent.

■

EXAMPLE 1.21 (Example 1.5-3, HT)

A red die and a white die are rolled. Let event C = {5 on red die} and event *D* = {sum of dice is ll}. Of the 36 equally likely outcomes, 6 are favorable to C, 2 are favorable to *D,* and 1 is favorable to $C \cap D$. Are the two events independent?

■

THEOREM 1.5 (Theorem 1.5-1, HT)

If *A* and *B* are independent events, then the following pairs of events are also independent:

(a) *A* and *B*'
(b) *A'* and *B*
(c) *A'* and *B'*

Proof:

■

EXAMPLE 1.22 (Example 1.5-4, HT)

An urn contains four balls numbered 1, 2, 3, and 4. One ball is to be drawn at random from the urn. Let the events *A, B,* and C be defined by $A = \{1, 2\}$, $B = \{1, 3\}$, and $C = \{1, 4\}$.

(a) Find P(A), P(B) and P(C).

(b) Find $P(A \cap B)$, $P(A \cap C)$, $P(B \cap C)$ and $P(A \cap B \cap C)$.

(c) Comment on results.

[independent in pairs – Pairwise independence] ■

DEFINITION 1.5 (Definition 1.5-2, HT)

Events *A*, *B*, and *C* are mutually independent if and only if the following two condition hold:

(a) *A*, *B*, and *C* are pairwise independent; that is,

$$P(A \cap B) = P(A)P(B), \quad P(A \cap C) = P(A)P(C), \quad (B \cap C) = P(B)P(C)$$

(b) $P(A \cap B \cap C) = P(A)P(B)P(C)$

Definition 1.5 can be extended to the mutual independence of four or more events. In such an extension, each pair, triple, quartet, and so on, must satisfy this type of multiplication rule.

If there is no possibility of misunderstanding, *independent* is often used without the modifier *mutually* when several events are considered.

EXAMPLE 1.23 (Example 1.5-6, HT)

A fair six-sided die is rolled six independent times. Let A_1 be the event that side i is observed on the ith roll, called a match on the ith trial, $i = 1, 2, \ldots, 6$, and let B denote the event that at least one match occurs.

(a) Find $P(A_i)$ and $P(A_i')$.

(b) Find $P(B)$.

■

EXAMPLE 1.24 (Example 1.5-7, HT)

The probability that a company's workforce has at least one accident during a certain month is $(0.001)k$, where k is the number of days in that month (say, February has 28 days). Assume that the numbers of accidents is independent from month to month. If the company's year starts with January, find the probability that the first accident is in April.

■

EXAMPLE 1.20 (Example 1.5-8, HT)

Three inspectors look at a critical component of product. Their responsibilities of detecting a defect are different, namely, 0.99, 0.98, and 0.96, respectively. If we assume independence, find the probability of at least one detecting a defect.

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Week 2 (SIT1001 Unit 1 Part 3 - 1 hour)

1.5 BAYES THEOREM

Bayes theorm can be used to calculate conditional probability, that is, how likely an event to occur related to any condition.

Other names used include Bayes rule or Bayes law.

We start the introduction to the Bayes Theorem using an example.

EXAMPLE 1.21 (Example 1.6-2, HT)

Three urns contains 6 red, 4 black; 4 red, 6 black; and 5 red, 5 black balls, respectively.

One of the urns is selected at random and a ball is drawn.

Let E_1= first urn is chosen

E_2= second urn is chosen

E_3= third urn is chosen

A = a red ball is drawn.

(a) State $P(E_1)$, $P(E_2)$ and $P(E_3)$.

(b) If the ball drawn is red, find the probability that is drawn from the first urn, that is, $P(A|E_1)$. Similarly, find $P(A|E_2)$ and $P(A|E_3)$.

(c) Given that the drawn is red, find the probability that it is drawn from the first urn, that is, $P(E_1|A)$.

■

Theory

Let B_1,B_2, ... , B_m constitute a *partition* of the sample space *S*. That is,

$$S = B_1 \cup B_2 \cup \cdots \cup B_m \text{ and } B_i \cap B_j = \emptyset,\ i \neq j.$$

The events B_1,B_2, ... , B_m are mutually exclusive and exhaustive (since the union of the disjoint sets equals the sample space *S*).

Futhermore, suppose the **prior probability** of the event B_i is positive; that is, $P(B_i) > 0$, $i = 1$, ... , m.

If A is an event, then A is the union of m mutually exclusive events, namely,

$$A = (B_1 \cap A) \cup (B_2 \cap A) \cup \cdots \cup (B_m \cap A).$$

Thus

$$\begin{aligned} P(A) &= \sum_{i=1}^{m} P(B_1 \cap A) \\ &= \sum_{i=1}^{m} P(B_1)P(A|B_i). \end{aligned}$$

Now, if $P(A) > 0$, then

$$P(B_k|A) = \frac{P(B_k \cap A)}{P(A)}, \qquad k = 1, 2, \ldots, m.$$

Hence, upon substitution, we have the **Bayes theorem**:

$$P(B_k|A) = \frac{P(B_k)P(A|B_k)}{\sum_{i=1}^{m} P(B_1)P(A|B_k)}, \qquad k = 1, 2, \ldots, m$$

The conditional probability $P(B_k|A)$ is often called the **posterior probability** of B_k.

Note: Bayes theorem is also called the formula for the probability of "causes".

Since the B_k's are a partition of *S*, one and only one of the event B_k occurs.

Hence, the above formula gives us the probability of a prticular B_k (i.e. a "cause") given that the event A has occured.

EXAMPLE 1.22 (byjus.com)

A man is known to speak the truth 2 out of 3 times. He throws a die and reports that the number obtained is a four. Find the probability that the number obtained is actually a four.

■

EXAMPLE 1.23 (Example 1.6-2, HT)

In a certain factory, machines I, II, and III are all producing springs of the same length. Of their production , machines I, II, and III, respectively produce 2%, 1%, and 3% defective springs. Of the total production of springs in the factory , machine I produces 35%, machine II produces 25%, and machine III produces 40%.

(a) If one spring is selected at random from the total springs produced in a day, find the probability that it is defective.

(b) If the selected springs is defective, find the conditional probability that it was produced by machine III.

Note how the posterior probability of III increased from the prior probability of III after the defective spring was observed, because III produces a larger percentage of defectives than do I and II. ■